



IEEE Transactions on Artificial Intelligence

Sponsoring Societies:



Submission Overview

Initial Submission

This manuscript has been submitted to the editorial office for review. Changes cannot be made during editorial review, but you can view the information and files you submitted, below.

Article Type	Research Article
Title	Adaptive- β Direct Preference Optimization for Stable Alignment of Large Language Models

Manuscript Files

Name	Type of File	Size
Main Manuscript.docx	Anonymized Main Document - MS Word	86.5 KB
Main Manuscript.pdf	Anonymized Main Document - PDF	364.1 KB
Title Page.docx	Title Page	22.7 KB

Abstract

Aligning large language models with human preferences is a central challenge in building reliable and trustworthy AI systems. Recently, Direct Preference Optimization (DPO) has been introduced as a simpler and more stable alternative to reinforcement learning from human feedback. Despite its structural advantages, DPO is highly sensitive to the choice of the preference scaling coefficient β , and using a fixed β can lead to training instability, excessive gradient growth, and undesirable drift from the reference policy. In this work, we propose Adaptive- β DPO, a novel approach in which the value of β is adjusted dynamically on a per-step basis using intrinsic training signals. These signals include the log-probability gap between preferred and non-preferred responses as an indicator of preference uncertainty, as well as a surrogate measure of Kullback–Leibler divergence to

control deviation from the reference policy. The proposed mechanism increases β for hard preference instances while reducing it when divergence grows, thereby balancing effective preference learning with training stability. Experiments conducted on the GPT-2 model using the Helpful-Harmless dataset demonstrate that, under standard DPO with a fixed β , increasing β from 0.05 to 0.30 leads to a higher final loss value, increasingly negative reward margins, and more than a sixfold increase in gradient norms, while reward accuracy remains nearly unchanged. This behavior highlights the strong sensitivity of DPO to manual β tuning. In contrast, Adaptive- β DPO improves training stability and prevents excessive preference pressure without modifying the original DPO loss formulation. These results indicate that adaptive β tuning can significantly reduce reliance on manual hyperparameter selection and provide a more controllable and interpretable framework for preference alignment in language models.

Authors	Title	Name	Email	Country/Location
	Mr.	Aref Ganjaee Sari Corresponding Author Submitting Author	aref.ganjaee1@yahoo.com	Iran
		 0009-0005-2988-6209		

Affiliations

Not provided

We will use the best match from our database to determine if your manuscript is eligible for special benefits. Matched organizations are for internal purposes and will not be published.

Funders	Is this research supported by funding? <i>No funding was received for this research</i>
---------	--

Additional Information

Keywords

- Adaptive algorithms
- Artificial intelligence safety
- Learning systems

- Machine learning
- Reinforcement learning
- Trustworthy artificial intelligence

Informed Consent

No, there were no human subjects

Animal Subjects

There were no animal subjects

Opposed Reviewers

Not provided

Cover letter / Comments

Yes, I'd like to add a cover letter or comments

Dear Editor-in-Chief and Associate Editors I am pleased to submit the manuscript entitled “Adaptive- β Direct Preference Optimization for Stable Alignment of Large Language Models” for consideration as a Research Article in IEEE Transactions on Artificial Intelligence. This manuscript addresses a fundamental limitation of Direct Preference Optimization (DPO), namely its sensitivity to the choice of a fixed preference strength parameter β . While DPO has emerged as a practical alternative to reinforcement learning from human feedback, existing approaches rely on manually selected, static β values, which can lead to unstable training dynamics, excessive gradient growth, and uncontrolled divergence from reference policies. In this work, we propose Adaptive- β DPO, a simple yet effective extension that dynamically adjusts β during training using intrinsic model signals reflecting preference uncertainty and deviation from the reference policy. Importantly, the proposed method preserves the original DPO loss structure and does not require reward models or reinforcement learning. Controlled experiments on the Helpful-Harmless dataset using GPT-2 demonstrate that Adaptive- β DPO significantly improves

training stability and gradient behavior compared to fixed- β baselines, without increasing computational complexity. We believe this contribution is well aligned with the scope of IEEE Transactions on Artificial Intelligence, particularly its focus on trustworthy, reliable, and theoretically grounded AI systems. The manuscript presents a clear methodological advance and offers practical insights for improving preference-based alignment pipelines. This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the submission and agree with its contents. Thank you for your time and consideration. We appreciate the opportunity to submit our work to your journal. Sincerely, Aref Ganjaee Sari

Has this manuscript been previously published, in whole or in part?

No, this manuscript has never been previously presented or published.

Do you have code (software) associated with your manuscript?

No, I do not have code associated with my manuscript

Impact Statement

This paper addresses a fundamental stability limitation in Direct Preference Optimization (DPO), a widely adopted alternative to reinforcement learning from human feedback for aligning large language models. While DPO simplifies alignment by removing explicit reward models, its performance and training stability critically depend on the choice of a fixed preference strength parameter β . Through controlled empirical analysis, we demonstrate that improper selection of β can lead to gradient explosion, excessive KL divergence, and unstable training dynamics without improving preference learning. To overcome this limitation, we introduce Adaptive- β DPO, a lightweight yet effective extension that dynamically adjusts the preference strength during training using intrinsic model signals. By jointly

accounting for preference uncertainty and deviation from the reference policy, the proposed method stabilizes optimization while preserving the original DPO objective structure. Unlike prior approaches that modify loss functions or introduce additional models, Adaptive- β DPO maintains simplicity and interpretability while improving training robustness. This work advances the understanding of preference-based alignment by reframing β as a dynamic control variable rather than a static hyperparameter. The proposed approach offers a practical and scalable solution for improving the reliability of alignment pipelines and provides a foundation for future research on adaptive control mechanisms in human preference learning.

Preferred Editors

Xiang Li

Do you have data associated with your manuscript?

No, I do not have data associated with my manuscript

History

Submitted On	4 January 2026 by Aref Ganjaee
	➤ Show this version history
Submission Started	4 January 2026 by Aref Ganjaee